

Program Question_4

```
implicit none
integer :: i, count
character(len=8) :: bit_str
character(len=8), dimension(0:255) :: all_bits
character(len=8), allocatable :: filtered_bits(:)

! Generate all 8-bit binary strings
do i = 0, 255
    write(bit_str, '(B8.8)') i
    all_bits(i) = bit_str
end do

print *, 'Ascending Order:'
do i = 0, 255
    print *, all_bits(i)
end do

print *, 'Descending Order:'
do i = 255, 0, -1
    print *, all_bits(i)
end do

! Filter out strings containing '101' or '100'
count = 0
allocate(filtered_bits(0:255)) ! Maximum possible
do i = 0, 255
    if (index(all_bits(i), '101') == 0 .and. index(all_bits(i), '100') == 0) then
        filtered_bits(count) = all_bits(i)
        count = count + 1
    end if
end do

print *, 'Number of valid bit strings (no 101 or 100):', count
print *, 'Filtered Bit Strings:'
do i = 0, count - 1
    print *, filtered_bits(i)
end do

deallocate(filtered_bits)

end program
```